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31 May 1983
SUPERSEDING
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MILITARY SPECIFICATION

BEAD BREAKER, PNEUMATIC TIRE, PORTABLE

(TIRE SIZES 14.00 BY 24 THROUGH 37.50 BY 33)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size of portable tire bead breaker for off-the-road equipment. The bead breaker facilitates tire removal while the wheel remains on the vehicle.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GGG-J-63	-	Jack, Hydraulic, Hand; Jack, Bumper (Hydraulic)
PPP-B-601	-	Boxes, Wood, Cleated Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	-	Box, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock Container Grade) and Cut Shapes
PPP-T-60	-	Tape, Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Materiel Readiness Command, ATTN: DRSAR-LEE-S, Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	-	Preservation, Methods of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-C-5501	-	Cap and Plug, Protective, Dust and Moisture Seal

STANDARDS

FEDERAL

FED-STD-H28	-	Screw Thread Standards For Federal Services

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking For Shipment and Storage
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one complete bead breaker for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of

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the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The bead breaker shall be designed to facilitate tire removal from off-the-road equipment that uses tires within a size range of 14.00 x 24 through 37.50 x 33. The bead breaker shall be designed to break both the inside and outside beads of the tire while the wheel is on the vehicle and also when the wheel is removed from the vehicle. The bead breaker shall produce pressure against the bead of the tire to break the tire loose from the rim flange and bead seat band. The breaker shall function on rims with or without bead loosening pockets. The bead breaker shall function on the rim without gouging or tearing the tire bead and sidewalls and without permanently deforming the rim beyond its initial manufactured state, though superficial scratches or dents that do not impair operation or cause air leakage are not considered to be deforming in nature. Total weight of the bead breaker shall not exceed 100 pounds.

3.2.1 Maintainability. Design and configuration of the bead breaker and the adjustment features shall insure that maintenance of operational capability and accuracy will be reduced to the simplest form, requiring a minimum of skills with standard maintenance equipment. Instructions for disassembly, repair, assembly, and adjustment shall be clear, concise, and definitive in application.

3.2.2 Interchangeability. All replaceable parts shall be manufactured to such standards that shall permit replacement or adjustment without the need for modification of the parts or the bead breaker.

3.2.3 Threads. All threaded parts of the bead breaker shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.3 Construction. All parts used in fabrication of the bead breaker shall be new and free of defects. In no event shall patching processes be used for reclaiming defective parts. Surfaces of all parts shall be clean and free of extraneous materials. External surfaces shall be free of sharp edges or irregularities.

3.3.1 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the tire equipment industry. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.3.2 Fastening and locking devices. All screws, pins, bolts, and similar locking parts shall be installed with means for adjustment to prevent loss of tightness. All parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

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3.4 Components and product characteristics.

3.4.1 Ram and pump assemblies. The ram and pump assemblies shall be essentially a remote control hydraulic jack conforming to type I, class 1, style A requirements of GGG-J-63. The assemblies shall have a rated capacity of not less than 10 tons. The ram assembly shall be self-retracting, shall have a minimum rise of 5.5 inches, and shall have a closed height of not more than 18 inches. Flexible tubing between the ram and pump assemblies shall not be less than 96 inches long. A bead breaking spade shall be attached or attachable to the plunger of the ram. The spade shall be designed to wedge between the tire bead and rim. The spade shall not ride up on the tire sidewall during operation.

3.4.2 Clamp head assembly. The clamp head assembly shall attach to the wheel rims of the tire size range specified herein to provide support of the ram assembly and spade during operation. The method of attachment shall have sufficient holding power to prevent the clamp head from backing off or slipping when pressure is applied to the head. The ram support shall be designed to allow the spade to wedge between the bead of the tire and rim and follow the rim contour during the pressing operation. The clamp head assembly shall be made of malleable iron or steel and shall not exceed 25 pounds.

3.4.3 Wedges. Not less than 6 wedges shall be furnished with each bead breaker. The wedges shall be of malleable iron or steel and shall have a configuration that permits spacing the bead away from the flange when the spade is removed.

3.5 Product identification. The bead breaker shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National Stock Number.

3.6 Painting. Unless otherwise specified (see 6.2), painting and finishing of the bead breaker shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall have been cleaned to be free of all foreign matter detrimental to painting, and at least one coat of primer and one coat of enamel, or equivalent, applied.

3.7 Workmanship. Standards of workmanship shall assure that the bead breaker shall have the strength, stability, and efficient operating characteristics found in commercial items and as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the product examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Performance test (see 4.6.1).
(c) Packaging inspection (see 4.7).

4.4.1 Sampling. When required (see 6.2), sampling for quality conformance inspection shall be in accordance with MIL-STD-105 (except for examination, as noted below) and the following:

Inspection/Test	Inspection Level	AQL
Product examination (see 4.5)	100% inspection	-
Performance test (see 4.6.1)	S-4	0 defects
Packaging inspection (see 4.7)	S-4	4.0

4.5 Product examination. Each complete bead breaker shall be examined to determine compliance with Section 3 requirements herein. As a minimum, the following characteristics shall be examined:

CHARACTERISTICS	REQUIREMENT PARAGRAPH
Design deficiencies	3.2
Maintainability not as specified	3.2.1
Interchangeability not as specified	3.2.2
Threads not as specified	3.2.3
Construction deficiencies	3.3
Fastening and locking devices not as specified	3.3.2
Components missing or not as specified	3.4 thru 3.4.3
Quantity of wedges not as specified	3.4.3
Product identification marking not as specified	3.5
Painting not as specified	3.6
Workmanship defects	3.7

4.6 Tests.

4.6.1 Performance test. The bead breaker shall be operated in accordance with the manufacturer's instructions to break the beads on both sides of tires not less than 29.50 x 29 (28 ply rating) in size.

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4.6.1.1 Performance test classification.

4.6.1.1.1 First article test. This test shall be performed on two tire and wheel assemblies for each bead breaker in the first article test sample. One tire and wheel assembly shall remain on the vehicle during the test. The beads on both sides of the tire shall be broken during the test. The second tire and wheel assembly shall be removed from the vehicle and then tested. During this test, the beads on both sides of the tire shall also be broken.

4.6.1.1.2 Quality conformance test. This test shall be performed on one tire and wheel assembly for each bead breaker in the test sample. The tire and wheel assembly may either remain mounted on the vehicle or be removed from the vehicle for performance of this test. The beads on both sides of the tire shall be broken during the test.

4.6.1.2 Performance test evaluation. Inability of the bead breaker to loosen the beads or evidence of any hydraulic fluid leakage during operation shall be cause for rejection. The tire shall be removed and examined for damage to the beads or sidewalls. The rim flange and the bead seat band shall be examined for deformation. The tire shall be remounted and inflated in accordance with the manufacturer's instructions and then examined for leaks due to damage or deformation caused by the bead breaker (see 3.2).

4.6.2 Ram and pump assembly test. The ram and pump shall be tested to determine compliance with 3.4.1.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging of the bead breakers to determine compliance with Section 5 requirements.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Industrial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Components of the bead breaker shall be cleaned and dried in accordance with any of the applicable procedures of MIL-P-116.

5.1.1.2 Preservative application. All exposed ferrous metal surfaces of the ram and pump assemblies that have not been painted or plated, shall be preserved by the application of type P-2 preservative. All other external unpainted, unplated ferrous metal surfaces shall be preserved with type P-1 preservative. The hydraulic fluid shall be shipped in the pump reservoir. The preservatives specified and the procedures of application shall be in accordance with MIL-P-116.

5.1.1.3 Unit packaging. The method of unit packaging shall conform to method I of MIL-P-116. Surfaces preserved with type P-2 preservative shall be covered with barrier material conforming to MIL-B-121, grade A, type and class

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optional. Secure barrier with tape conforming to PPP-T-60, type optional. The flexible tubing between the ram and pump assemblies may be disassembled from the units provided the ends of the tubing and apertures of the assemblies are completely sealed with plastic caps or plugs conforming to MIL-C-5501 or with tape conforming to PPP-T-60, type IV. In the event the tubing is disassembled from the assemblies, it shall be coiled to a minimum safe diameter and secured with cotton tape or twine to prevent uncoiling. The bead breaker shall be protected from damage by wrapping each component with cushioning material conforming to PPP-C-843, type II or by insulating components, utilizing die-cut fiberboard conforming to PPP-F-320, class weather-resistant. The unit container shall conform to PPP-B-636, type CF, class weather-resistant. Technical publications shall be packaged in accordance with method IC-1 of MIL-P-116 and placed in the unit container.

5.1.2 Industrial. The bead breakers shall be preserved and packaged as specified in ASTM D3951-82.

5.2 Packing. Packing shall be Level A or Industrial, as specified (see 6.2).

5.2.1 Level A. The packaged bead breaker shall be packed in an exterior shipping container conforming to PPP-B-621, class 2, or PPP-B-601, overseas type. The gross weight of each shipping container shall not exceed the weight limitation of the applicable container specification. Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. Closure and strapping of the exterior container shall be in accordance with the appendix to the applicable container specification.

5.2.2 Industrial. Industrial packing of the bead breakers shall be as specified in ASTM D3951-82.

5.3. Marking.

5.3.1 Level A. Interior packages and exterior shipping containers shall be marked as specified in MIL-STD-129.

5.3.2 Industrial. Interior packages and exterior shipping containers shall be marked as specified in ASTM D3951-82.

6. NOTES

6.1 Intended use. The bead breaker covered by this specification is used for separating tire beads from wheel rims prior to tire removal from large off-the-road equipment. The bead breaker is intended to be used with the wheel and tire assembly either on or off the vehicle.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Identification marking if different (see 3.5).
(d) Painting if different (see 3.6).
(e) First article inspection if different (see 4.3).

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(f) Quality conformance inspection if different (see 4.4).
(g) Sampling when required (see 4.4.1).
(h) Level of preservation, packing, and marking required (see 5.1, 5.2, and 5.3).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment complies with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - AL
Navy - YD

Preparing activity:
Army - AL

Project No. 4910-0540

Review activities:

Army - SM
DLA - CS

User activities:

Army - ME
Navy - MC

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